

EXPERIMENT 5(A) - BUILDING AND VALIDATING LINEAR MODELS

UCI DIABETES DATASET

	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	182	68	35	285	28.16	
1	131	102	41	96	28.60	
2	172	89	15	283	32.29	
3	94	116	41	146	35.05	
4	186	94	13	3	45.94	

	DiabetesPedigreeFunction	Age	Outcome
0	1.800	74	0
1	1.256	74	1
2	1.007	23	0
3	1.792	79	0
4	0.697	70	0

UCI Dataset Shape:
(100, 8)

PIMA INDIANS DIABETES DATASET

	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	88	65	44	80	43.58	
1	164	115	42	121	41.76	
2	147	58	19	34	45.62	
3	183	103	44	209	37.61	
4	167	107	13	16	26.74	

	DiabetesPedigreeFunction	Age	Outcome
0	0.749	22	0
1	0.684	79	0
2	1.125	36	1
3	1.725	75	1
4	0.629	46	1

Pima Dataset Shape:
(100, 8)

FEATURES AND TARGET

Independent Variables:

- Glucose
- BloodPressure
- BMI

Dependent Variable:

- Age

TRAIN / TEST SPLIT

UCI Training samples : 80
UCI Testing samples : 20

Pima Training samples : 80
Pima Testing samples : 20

MODEL TRAINING

UCI Linear Regression Model trained successfully.
Pima Linear Regression Model trained successfully.

UCI DIABETES DATASET - LINEAR REGRESSION RESULTS

R² Score : -0.1077
MSE : 335.4475
MAE : 16.0215

PIMA INDIANS DIABETES DATASET - LINEAR REGRESSION RESULTS

R² Score : 0.0514
MSE : 275.7978
MAE : 13.8539

UCI ACTUAL VS PREDICTED

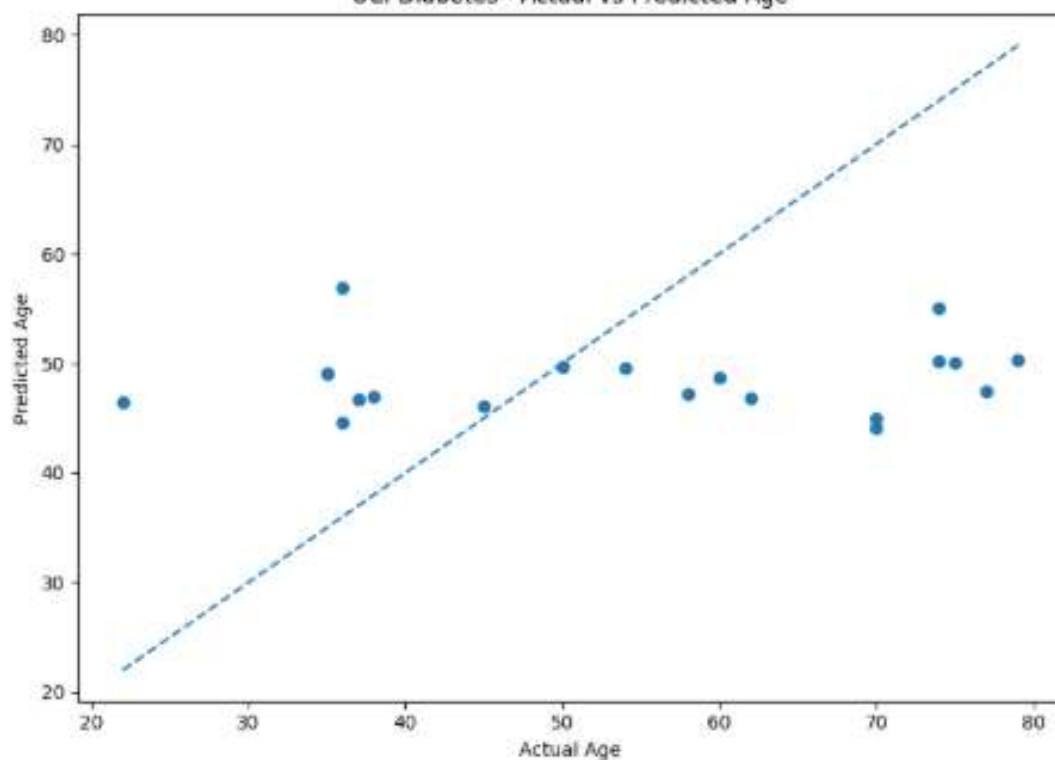
Actual Age	Predicted Age
45	46.05
37	46.48

Actual Age	Predicted Age
45	46.05
37	46.68
77	47.48
62	46.88
75	50.09
58	47.16
22	46.42
68	48.74
50	49.75
74	50.23
35	49.06
54	49.59
36	44.61
35	49.02
70	44.93
70	44.09
74	55.06
36	56.91
79	50.30
38	46.98

PIMA ACTUAL VS PREDICTED

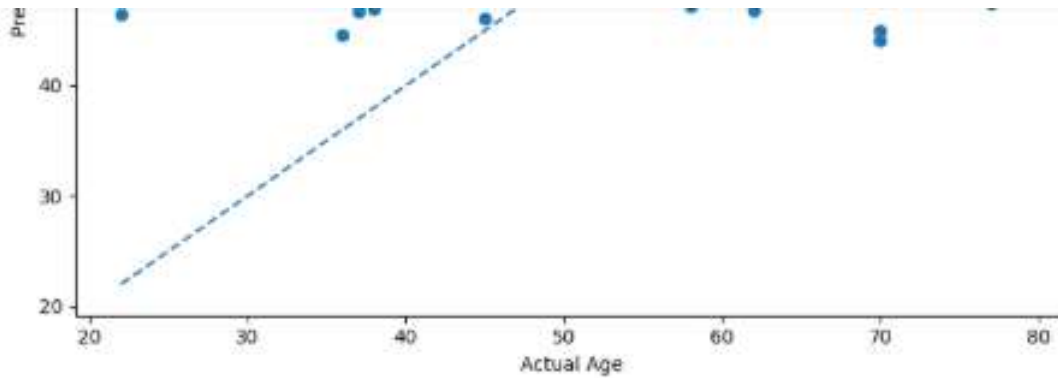
Actual Age	Predicted Age
43	54.82
72	51.10
50	50.06
59	46.51
54	51.49
22	45.41
29	45.57
43	51.75
41	48.97
22	50.78
58	56.75
44	48.66
28	46.66
22	51.63
70	57.74
46	52.00
35	49.66
79	51.24
29	52.03
63	57.08

UCI Diabetes - Actual vs Predicted Age

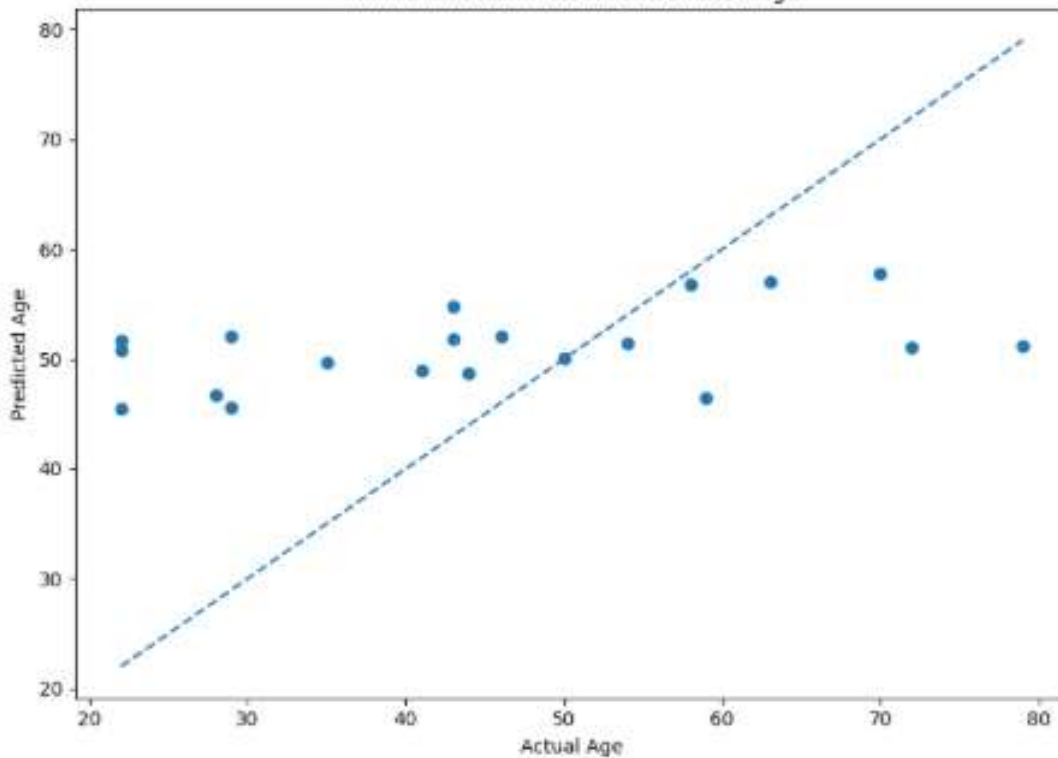


Pima Diabetes - Actual vs Predicted Age





Pima Diabetes - Actual vs Predicted Age



FINAL MODEL COMPARISON

Dataset	R2 Score	MSE	MAE
UCI Diabetes	-0.1077	335.4475	16.8215
Pima Indians Diabetes	0.0514	275.7978	13.8539

RESULT

Linear Regression models were successfully built and validated for both the UCI Diabetes and Pima Indians Diabetes datasets. Glucose, BloodPressure, and BMI were used as independent variables. Age was used as the target variable. The models were evaluated using R² Score, Mean Squared Error, and Mean Absolute Error. Actual versus predicted values were visualized.

EXPERIMENT 5(A) COMPLETED SUCCESSFULLY